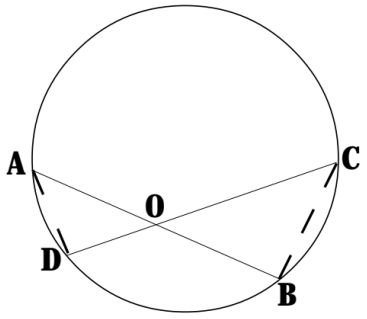
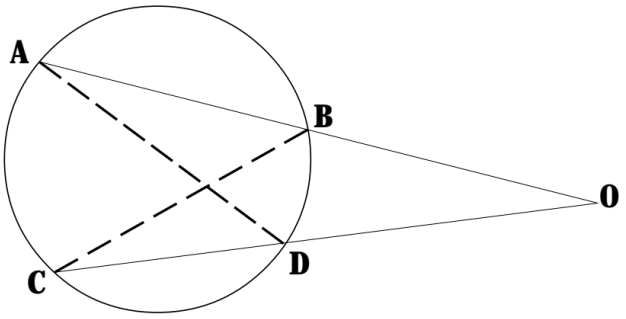
7.Rectangles in connection with circles

Theorem

If any two chords of a circle cut one another internally or externally, the rectangle obtained by the segments of one is equal to the rectangle obtained by the segments of the other.



(i)



(ii)

Given : **Let the chords AB, CD cut one another at O, [internally in figure (i), externally in figure (ii)]**

To prove : **OA . OB = OC . OD**

Construction : **Join AD, BC.**

Proof : **In triangles AOD, COB.**

$\angle$OAD = $\angle$OCB (Angles in the same segment)

$\angle$AOD = $\angle$COB (Vertically opposite angles)

Therefore, third angles in each are equal.

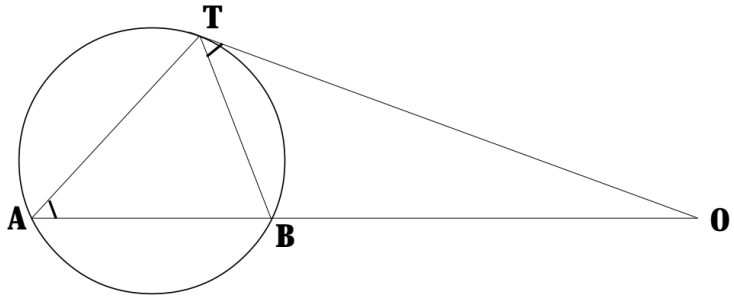
Hence AOD, COB triangles are similar

$$\frac{AO}{CO}=\frac{OD}{OB}$$

AO . OB = CO . OD

Co Theorem

If from an external point a secant and tangent are drawn to a circle, the rectangle contained by the whole secant and the part of it outside the circle is equal to the square on the tangent.



Let OBA be a secant and OT a tangent drawn to the circle from the point O.

Given : **OBA is a secant and OT is a tangent.**

To Prove : **$OA \cdot OB = OT^2$**

Construction : Join BT, AT

Proof : In triangles OAT, OTB.

$\angle AOT = \angle BOT$

$\angle OAT = \angle OTB$ (angles in alternate segment)

Therefore third angle in each are equal

Hence OAT, OTB triangles are similar

$$\frac{OA}{OT} = \frac{OT}{OB}$$

$OA \cdot OB = OT^2$

Exercise 7

1 **O is the centre of a circle of radius 6 cm. A secant PXY drawn from a point P outside the circle meets the circle at X and Y. If OP = 10 cm find the length of the tangent drawn from the point P to the circle. If PX = 5 cm, find the length of XY.**

2 **A tangent drawn to a circle from a point P is PT. PQR is a secant to the circle. If PQ = 4 cm and PT = 8 cm, find the length of QR.**

3 In an acute angled triangle ABC, altitudes BD and CE intersect at H
Prove that BH : HD = EH : HC

4 The length of a bridge which connects banks of a river is 100m A foot path designed above the bridge looks like an arc of a circle. Two pillars A and B at the end points of the bridge bear the foot path.
If the highest point C in the foot path is 20 m from the bridge, find the radius of the arc.

5 TA and TB are two tangents drawn from a point T to a circle. OT intersects AB at X.
Prove that (i) AX . XB = OX . XT
(ii) OX . OT = OA^2

6. A, B are centres of two circles, C_1 and C_2, which do not intersect each other. A third circle with centre O intersect circle C_1 at C, D and C_2 at E, F.
If two straight lines CD, EF produced to meet at P, prove that the lengths of the tangents drawn from P to C_1 and C_2 are equal.

7. AB and AC are two chords of a circle. A straight line parallel to the tangent at A, intersects AB and AC at D and E respectively.
Prove that AB . AD = AC . CE.

8. PQ and PR are two chords of a circle. Another chord PS of this circle intersects QR at T.
Prove that PS . PT = PQ^2